Cd CIA - 02 "closed loop analysis of buck converter using dynamic load conditions"

01. Buck converter Transfer function "use control system"
02. look online ; EV Battery profiles [voltage / power]
03. Take png or Jpg files V & P profiles
04. Go to web plot digitizer ["use youtube video to know web plot digitizer]
05. Make buck convert in plecs → use V & P profiles specifications - also import files we got
06. switching freq = 100kHz
07. Calculate L for 20% ripple
08. Plecs → Refer / closed loop Transfer function
09. vary duty ratio
10. Take Results.

Dear Students,

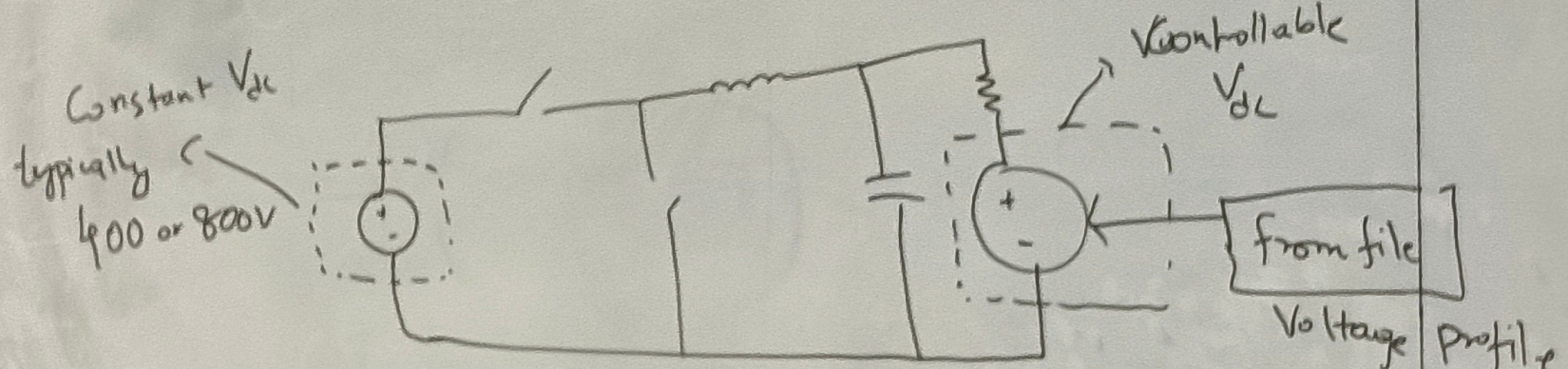
I am making your life easy. Please find the attached file where you can find the voltage profiles of different EVs.

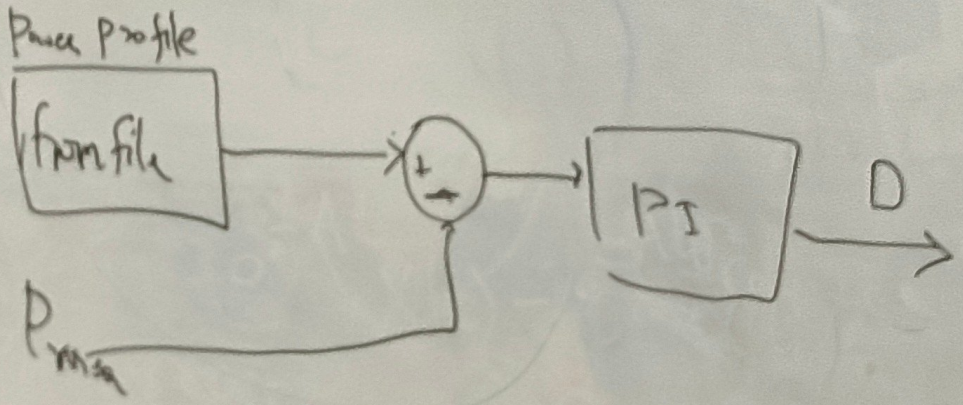
Use the webplotdigitizer to extract the data from the raw profiles: WebPlotDigitizer - Copyright 2010-2024 Ankit Rohatgi

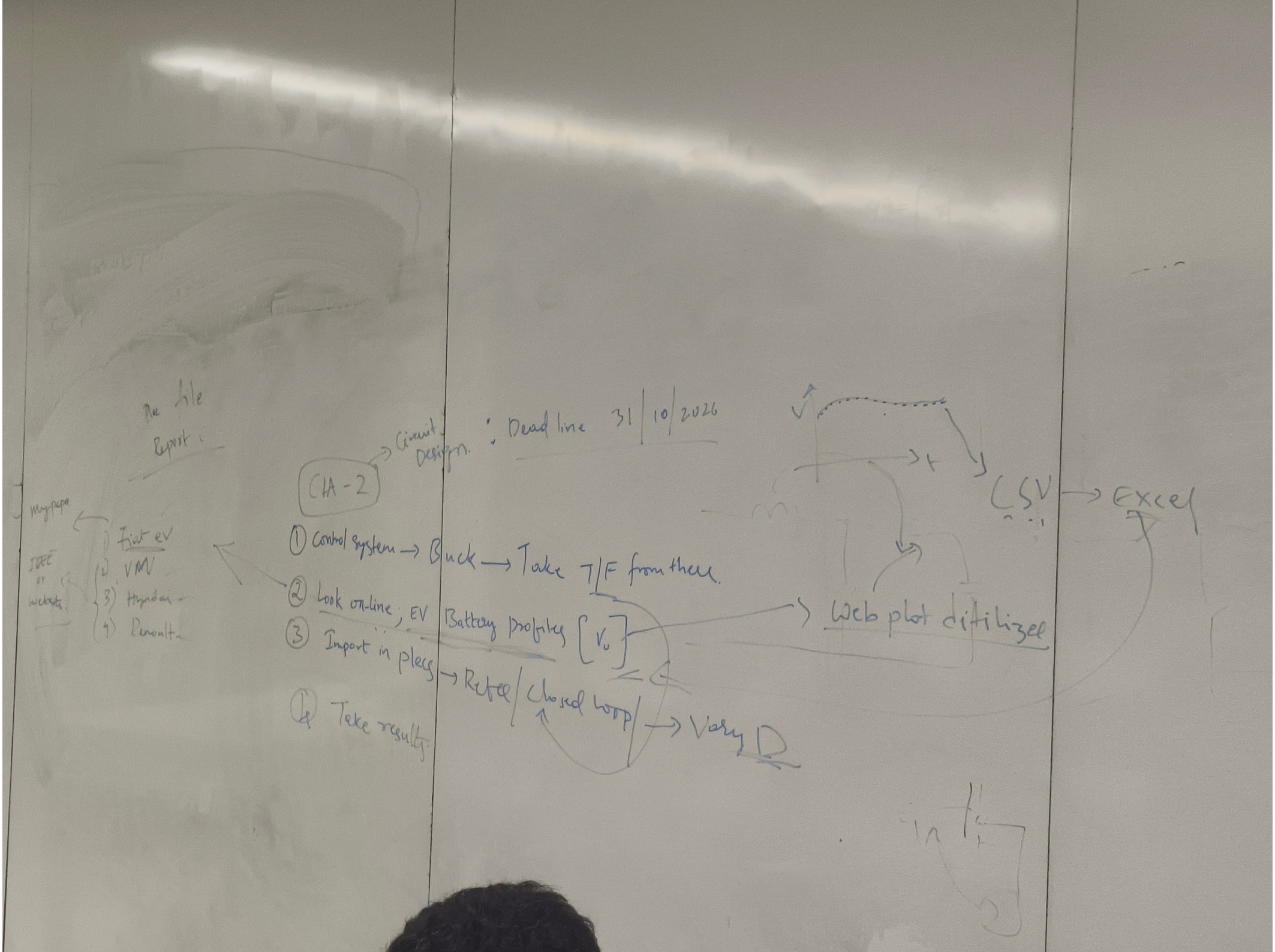
Follow the youtube video to better understand the procedure on how to use this tool: How to extract Data from figures to Excel: WebPlotDigitizer- Full Easy Tutorial. #dataextractor

Best Regards,

① Voltage profile
② Power profile
③ Specifications
Same power & Voltage
④ Switching freq = 100kHz
⑤ Calculate L for 20% ripple.






Report :
CIA - 2
Circuit Design.
: Deadline 31/10/2026
1) Control system → Buck → Take T/F from there.
2) Look on-line ; EV Battery profiles [V₀]
3) Import in plecs → Refer / Closed loop / → Vary D
4) Take results.
1) Fiat ev
2) VMV
3) Hyndai
4) Renault
CSV → Excel
Web plot digitizer